

Learning goals: Basics of shell scripting continued: for, while, until loops, functions

NOTE: Please do all your work in a new folder called `lab12`. Save the script for each exercise separately as `question1.sh`, `question2.sh` and so on. For all questions with functions, please store the output of the function in a variable and then print the value of that variable in the main script. At the end of the lab, please ‘push’ your scripts to GitHub!

EXERCISES

- (1) Write a shell script to print numbers from 0 to 10.
- (2) Write a shell script to print only even numbers between 0 and 50. (Note: think about what is the condition to be tested as well as the for-loop structure.)
- (3) Write a shell script to make the user input a number and making use of the `until` loop, print the multiplication table of that number up to 15.
- (4) Create a simple text file called `nums.txt` with few numbers in the same line, for example `2 3 5 7`. Then read these numbers in a shell script into a variable array called `numbers` using the following command: `read -ra numbers < nums.txt` . Print the elements of this variable array, and using a `for` loop, print the result of doubling of each number in the array.
- (5) Write a bash script that defines a function called `divide` that takes two numbers as arguments and prints their quotient and remainder. Make sure to keep all the variables within the function as local variables and also prints the result up to 2 decimal places. This function also should check and make sure that division by zero does not happen.
- (6) Write a bash script that defines a function called `maximum` that takes two numbers as arguments and prints the maximum of the two.
- (7) Write a function to check if a directory exists, and if it does not exist, to create the directory. If the directory exists, then the function should list the files in the directory.